PrismaQuant and PrismaSCOUT: Operational Rate–Distortion for Mixed-Precision Large Language Model Quantization

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Abstract

We describe PrismaQuant, a mixed-precision quantization pipeline for large language models, and PrismaSCOUT, the search-and-select algorithm at its core. Mixed-precision allocation is conventionally posed as a constrained discrete optimization in which a per-Linear surrogate cost $\sum_i u_i(f_i)$ is minimized over format assignments under a total-bit budget. We argue that this formulation is structurally biased on modern transformers: the true joint quantization error is non-additive across layers, and surrogate ranks reliably overshoot when many flips are committed at once. We therefore decouple *candidate generation* from *candidate selection*. Surrogate costs propose discrete allocations cheaply via a multi-level hierarchy (L_1 probe, L_2 perturbed-activation fixed point, L_3 propagated end-KL); every assignment that may ship is then gated by a real, end-to-end Kullback–Leibler divergence measured on a held-out calibration split. Selection is performed by a validated-frontier kneedle: the bisection samples (bpp, KL) pairs adaptively, the dominance filter is widened by an explicit KL noise floor, and the chosen knee is verified by a leave-one-out stability diagnostic. A monotone coordinate-descent polish guarantees the shipped assignment is no worse than the chosen frontier point. On Qwen3.6-27B, the pipeline produces a 5.31 bits-per-parameter artifact with held-out KL 0.0151, replacing a previously shipped 5.50 bpp artifact at KL 0.0475 from the same source weights and the same export-time quantization tricks — 11% smaller and 68% lower KL on the same calibration. The artifact is published with DOI `10.57967/hf/8656` [1]. We also document, in a separate methodology section, several numerical methods that were proposed and either implemented-but-deprecated or rejected outright during the design process: Lagrangian λ-bisection on the dual gradient, sandwich proximal recalibration, block-DP over architectural cliques, sparse pairwise QUBO refinement, top-K Hessian-eigenvalue covering, surrogate-only knee selection, and a single-anchor probe-only knee predictor. We give the principled argument for each, and the empirical or structural reason it did not become the shipping path.

1 Introduction

Post-training quantization has converged on two dominant axes. Per-tensor methods — GPTQ [4], AWQ [5], AutoRound [6], HALO/HQQ-style rotation preprocessors, and most recently ParoQuant [15] — improve the per-Linear quantization error of a fixed format. Allocation methods — HAWQ-V3 [8], the cross-layer-dependency-aware integer-quadratic-programming approach of CLADO [9], the cooperative-game CoopQ allocator [10], and the AutoML-style AMQ [11] — choose how many bits to spend where. The two axes are largely orthogonal: an allocator can sit on top of any per-tensor toolkit and decide which Linears get 16, 8, 4, 3, or 2 bits.

The allocation literature is unified by a recurring observation: per-layer sensitivity surrogates (diagonal Hessian traces, Fisher MSEs, output-MSEs) are biased estimators of joint quantization error. The standard remedy is to model the bias — pairwise integer-quadratic programming over an explicit cross-layer dependency model in CLADO [9], second-order ILP in HAWQ-V3 [8], Shapley-based cooperative-game allocation in CoopQ [10]. We take a different tack. *Surrogates generate, real measurement selects.* An allocator does not need a perfect cost model if every candidate it proposes can be cheaply re-scored end-to-end on a held-out calibration split. Cross-layer interactions and propagated effects then cease to be quantities one must *model*; they are quantities one *observes*, automatically, every time a candidate is gated.

This paper reports the design and engineering of a system, named PrismaQuant, that operationalizes that principle, and the search algorithm at its core, PrismaSCOUT (**S**urrogate-**C**ascaded **O**ptimization **U**nder **T**radeoff). Three contributions are new:

1. A **multi-level cost hierarchy** (L_1, L_2, L_3) that separates cheap proposal from increasingly faithful measurement. L_2 is a perturbed-activation fixed point that folds inter-layer coupling into the activation cache rather than into a closed-form interaction term. L_3 is a propagated end-KL measurement performed against a target-specific BF16 paired baseline, lane-batched and tail-amortized so that one L_3 pass at Qwen-27B scale takes a tractable wall time.
2. A **validated-frontier kneedle**: the elbow of the operational rate–distortion curve is found on *measured* (bpp, KL) pairs, not on a surrogate hull. We add a KL noise floor to the dominance filter, an adaptive bisection that refines around the current best knee, a leave-one-out stability diagnostic, and a strict monotone polish.
3. A documented **methodology of rejected detours**. Several numerically attractive ideas (sandwich recalibration, λ-bisection, block-DP, sparse pairwise QUBO, top-K Hessian covering, surrogate-only knee, probe-only prediction) were proposed during design, some implemented; we report the conditions under which each is principled and the empirical reasons each was not adopted as the shipping path.

Headline empirical result. On Qwen3.6-27B, with identical source weights and identical export-time per-tensor tricks (HALO, GPTQ damp sweep, scale sweep, block-output match), PrismaSCOUT produces a 5.31 bpp artifact with held-out KL 0.0151. The previously published PrismaQuant v1 artifact at the same source was 5.50 bpp with held-out KL 0.0475 from the same calibration. The new artifact is 11% smaller and the held-out KL is 68% lower. Only the selection routine changed.

The paper is organized as follows. Section 2 gives background and related work. Section 3 sets up the theoretical framework: the rate–distortion view, the additive surrogate failure mode, the cost hierarchy, the kneedle. Section 4 states the algorithm. Section 5 is a deliberate methodology section on the numerical detours we considered and rejected. Section 6 reports the engineering work that made the measurement loop tractable. Section 7 reports empirical results on Qwen3.6-4B and Qwen3.6-27B, including an honest accounting of the tool-eval-bench delta. Section 8 discusses limitations.

2 Background and Related Work

Per-tensor quantization. GPTQ [4] is the workhorse weight-only quantizer for transformers, posed as Hessian-weighted least squares over reconstruction error. AWQ [5] reweights salient channels. SmoothQuant [7] pushes activation quantization tractable by migrating activation magnitude into weights. AutoRound [6] sign-rounds via signSGD. HALO/HQQ-style rotations preprocess weight matrices to reduce outlier counts before integer quantization. These methods take a single matrix and a fixed format and produce a high-quality quantized version of that matrix; they do not decide *which* matrix gets which format.

Mixed-precision allocation. HAWQ-V3 [8] formulates mixed-precision quantization as an integer linear program over per-layer sensitivity (second-order Hessian information) and dyadic arithmetic constraints. CLADO [9] extends this with an explicit cross-layer dependency: pairwise quantization-error coupling is measured on a small data subset and the bit-allocation problem is solved as an integer quadratic program in seconds, rather than approximated additively across layers. Recent allocation work for LLMs includes CoopQ [10], which models layers as a cooperative game and uses a Shapley-based progressive allocator to capture inter-layer interaction, and AMQ [11], which applies AutoML-style search over per-layer weight-only bit-widths. Surveys of the broader mixed-precision space have appeared as the field has matured [12], and recent iterative allocators include IterQuant [13]. The geometric view of quantization error in [14] casts GPTQ as Babai's nearest plane algorithm and is orthogonal to allocation but informs how a fixed allocation should round.

Knee detection and operational rate–distortion. The kneedle algorithm of [16] provides a non-parametric estimator of the elbow of a tradeoff curve. The closely related operational rate–distortion (ORD) framing of [17] establishes that any discrete-choice rate–distortion problem admits a Lagrangian relaxation whose solutions trace the lower convex hull of the achievable points. Knees on the ORD frontier correspond to phase transitions where additional rate buys disproportionate fidelity. Our work imports the ORD vocabulary — and avoids the convex hull, since for mixed-precision LLM quantization the discrete frontier admits non-convex segments and a λ-sweep on the dual would skip them (Section 5.1).

Microscaling formats. The MXFP4/MXFP6/MXFP8 family is specified by the OCP microscaling working group [18]: 4/6/8-bit elements with shared FP8 (E8M0 or E5M2) block scales over groups of 32. NVFP4 is NVIDIA's narrower-group 4-bit format with 16-element FP8 group scales [19]. NVFP4 weight-only Linears cost $4+8/16 = 4.5$ bpp; MXFP4 weight-only Linears cost $4 + 8/32 = 4.25$ bpp. Modern Blackwell/Hopper GPUs offer first-class kernels for both. Mixed precision in this paper means an assignment $\mathcal{A} : \mathcal{L} \to \mathcal{F}$ where each Linear independently selects from $\{$BF16, MXFP8, NVFP4, MXFP4, NVINT3, $\dots\}$, subject to fused-sibling constraints in attention (q, k, v) and MoE $(\text{gate\_up}, \text{down})$.

Serving. The artifacts described here are exported to vLLM [20] via the `compressed-tensors` schema, with format coverage extending to NVFP4, MXFP8, BF16 passthrough, and (in MoE settings) MXFP4 experts. End-to-end inference uses Blackwell-native kernels.

2.1 The PrismaQuant v1 baseline and why it sets a hard bar

The artifact PrismaSCOUT replaces was not a weak baseline. The production PrismaQuant v1 stack is a tuned per-tensor pipeline that combines several state-of-the-art techniques per Linear, with a per-layer sensitivity allocator on top. It is worth describing in its own right, both because PrismaSCOUT inherits it wholesale (only the selection routine changed in the head-to-head comparison of Table 1), and because understanding what v1 already captures clarifies which residual is left for joint-allocation methods to attack.

The per-tensor toolkit. For every quantizable Linear, v1 composes:

- **HALO rotation preprocessing** [3] — a Hadamard-style orthogonal rotation applied to the weight matrix before integer quantization, which spreads outlier-channel mass over the rotated basis and reduces the per-element representation range. Particularly effective for 4-bit weight-only formats whose error is dominated by tail outliers.

- **Per-layer GPTQ** [4], the workhorse Hessian-weighted least-squares rounding. v1 ships an adaptive *damp probe* (commit `0cbaddb`) that tries multiple Hessian damping coefficients per Linear and picks the lowest measured output MSE; a *batched-NVFP4* fast path (commit `f57d960`) processes Linear groups together.
- **Scale sweep** (commit `7884536`) — a per-tensor scale-factor search that minimizes the post-quantization output MSE of each Linear, executed inside the export hot path.
- **Block-output matching** (commits `9ef0a05`, `a6c0cf6`, `b12f96b`) — after per-Linear quantization, a block-level optimization that re-tunes a subset of parameters so that the *block* output matches the BF16 reference, not just the individual Linear outputs.
- **FP8 source passthrough** (commits `705fbcd`, `0bf4146`) — for natively-FP8 source checkpoints, the allocator may pick a verbatim copy of the source FP8 codes plus `weight_scale_inv`; the BF16 view is a lossless dequant, so the per-Linear Δloss is exactly zero. This is important for native-FP8 architectures (MiniMax, DeepSeek-class).
- **Activation clipping and FP32 activation cache** (commit `a5d13db`) — wins the cost-step vs. activation quantization tradeoff for formats that benefit from clipped activations.

The allocator on top is HAWQ-style per-layer sensitivity [8] fed by a multi-chunk empirical-Fisher probe, with a multi-choice knapsack DP under the bit budget. Activation residency, fused-sibling coherence, shape constraints (e.g. MXFP8 tile alignment, FlashInfer kernel masks), and joint-input-global-scale handling for fused attention are all enforced inside the DP. The whole pipeline runs end-to-end on a UMA host with bounded memory.

Why this works as well as it does. Per-layer quantization error is, on most Linears in a modern transformer, well-modeled by a local quadratic $\frac{1}{2}\langle H_i, (W_i - \hat{W}_i)^2\rangle$. Each component above attacks one term of this quadratic: HALO reduces $\|W_i - \hat{W}_i\|$ by spreading outliers; GPTQ minimizes $\langle H_i, (W_i - \hat{W}_i)^2\rangle$ exactly under a fixed format; scale sweep tunes the post-rotation per-tensor scale to minimize the remaining error; block-output match closes the gap to the *block-level* reference, absorbing some of the inter-Linear coupling without an explicit cross-layer cost model. The allocator's HAWQ-style sensitivity then spends bits on the Linears whose local quadratic has the largest top-eigenvalue. Each per-layer technique is narrow, principled, and composable; together they extract most of the per-Linear quantization headroom.

The empirical evidence that this pipeline is strong is the community adoption: the public Qwen-3.6 27B, Qwen-3.5 122B-A10B, Qwen-3.6 35B-A3B, Mistral-Medium-3.5, MiniMax-M2.7, Gemma-4-31B, and DSv4-Flash PrismaQuant v1 artifacts have together accumulated over 60,000 downloads on Hugging Face in their first weeks of release, across a heterogeneous set of source architectures (dense, MoE, and native-FP8). v1 is a competitive shipping default in absolute terms.

The stacking-interference failure mode. The conventional intuition for per-tensor techniques is that they are *additive* in single-format quantization: HALO reduces outliers, GPTQ minimizes the local quadratic, scale-sweep tunes the post-rotation per-tensor scale, block-output-match closes the residual block-level gap, and stacking them composes the gains. In a fixed-format setting (e.g. all NVFP4) this composition is empirically robust on the per-Linear metrics each pass optimizes against. In a mixed-precision setting it is not. Each pass operates on the activation distribution as modified by the previous pass, and the per-Linear cost models the passes consume are decoupled from the format choices that the allocator will eventually make for upstream Linears. The result is that pairs of individually validated wins can interfere catastrophically when stacked.

The canonical empirical incident is recorded in the quality-wins audit on Qwen3-0.6B (handover `.claude/codex-handover-2026-04-29-pm.md`). Each of four wins (PRISMAQUANT_ACT_CLIP_QUANTILE=0.999, PRISMAQUANT_GPTQ_DAMP_SWEEP, PRISMAQUANT_BLOCK_OUTPUT_MATCH, PRISMAQUANT_DO_NO_HARM) was validated individually on the PRISMAQUANT validator EVAL_PROMPTS suite. solo_clip alone moved PPL from 15.453 (baseline) to 14.546 (-0.907); the other three were neutral or near-neutral on Qwen3. The cumulative full_v2 configuration with all four wins enabled landed at PPL 14.987, -0.466 from baseline — materially *worse* than solo_clip alone. Roughly half the activation-clip benefit was lost. The mechanism, root-caused after the audit: the do-no-harm gate compares per-Linear MSE on *unclipped* cached activations while GPTQ optimized that Linear's weight under *clipped* activations. The post-pass weight is a better fit on the clipped activation distribution but appears worse on the unclipped one; the gate reverts clip-improved Linears back to RTN, undoing the upstream gain. This is exactly the AWQ-style interference pattern, and it is structural. Mixed-precision quantization sits on a directed acyclic graph of optimizations whose intermediate states are not propagated forward through the rest of the pipeline.

A second instance of the same mechanism is the AWQ result: AWQ [5] was defaulted off in v1 after empirical bakeoffs showed it *worsened* NVFP4 group quantization on most Linears, despite being a strict win on uniform-format 4-bit settings. The per-Linear "do no harm" gate (commit ed9c60c) was added as a direct band-aid for the same family: after GPTQ + scale-sweep, compare against pure RTN on the cached activations and revert that Linear if RTN beats the post-pass weight. Both observations are artifacts of the same structural property — per-tensor wins that modify the activation distribution do not propagate that modification forward into the cost models the downstream wins consume, so cumulative recipes are not, in general, the composition of their individually-validated parts.

What v1 leaves on the table. The systematic gap v1 cannot close from the per-layer view is the joint-allocation question: "which Linear should pay an extra bit?" Per-layer sensitivity correlates with end-KL impact but does not predict it perfectly, because end-KL depends on how a Linear's quantization error *propagates* downstream, and propagation is sensitive to the formats chosen for upstream Linears. v1's HAWQ-style allocator overspends on Linears with high local sensitivity but modest downstream impact, and underspends on Linears with low local sensitivity but errors that compound through later attention. The stacking-interference failure mode of the previous paragraph is the mirror image of this gap on the per-tensor side: when upstream optimization decisions are not propagated into the cost model that downstream optimizations consume, the resulting cumulative recipe is not the composition of its individually-validated parts. PrismaSCOUT does not change a single per-Linear technique; it changes the allocator from *additive surrogate cost* to *validated end-KL on real assignments*, and it gates every shipping decision on a held-out KL measurement. The real-KL gate is also what catches the stacking-interference failure mode end-to-end — if a recipe stack quietly regresses on real KL, no DP rearrangement of the per-Linear costs hides that fact. The Qwen3.6-27B head-to-head in Table 1 is the empirical statement of this relationship: the per-Linear toolkit is held fixed, only the selection routine changes, and the resulting artifact is 11% smaller and 68% lower in held-out KL. The remaining headroom is at the joint level, which is where the rest of this paper goes.

3 Theoretical Framework

3.1 The shipping criterion

Let p_θ denote the full-precision teacher model with parameters θ and $p_{\hat{\theta}}$ a quantized variant with weights $\hat{\theta} = Q_{\mathcal{A}}(\theta)$ under format assignment $\mathcal{A}$. Given a calibration distribution $\mathcal{D}$ over input prefixes $x_{1:t}$, the operational quality of $\mathcal{A}$ is

$$D_{\mathrm{KL}}(\mathcal{A}) \;=\; \mathbb{E}_{x\sim\mathcal{D}}\, D_{\mathrm{KL}}\big[\, p_\theta(\,\cdot \mid x) \,\big\|\, p_{\hat{\theta}}(\,\cdot \mid x) \,\big]\,. \tag{1}$$

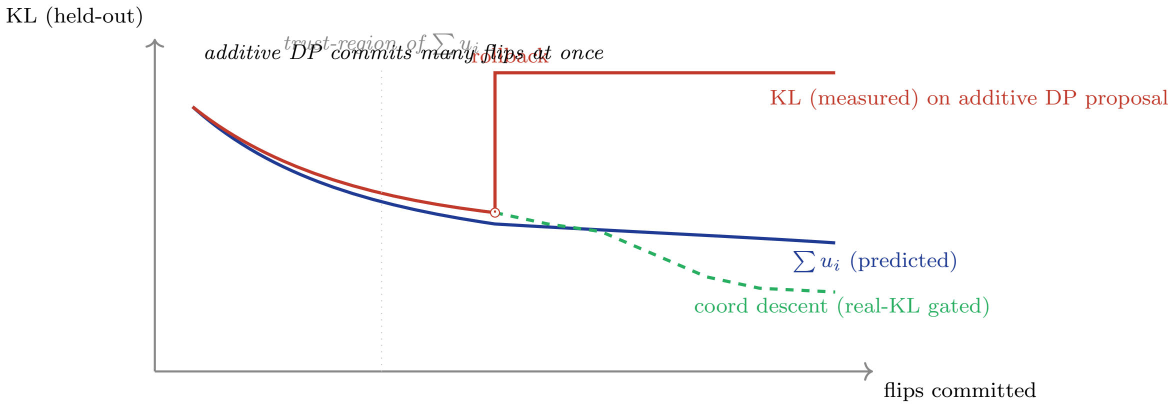

Figure 1: The non-additivity overshoot. The additive surrogate $U_{\text{add}} = \sum u_i$ predicts a monotone KL decrease as flips accumulate (blue). The real, measured held-out KL agrees with the surrogate inside the trust region of the local Taylor expansion, but diverges sharply when many flips are committed at once (red). The validated-frontier gate detects the regression and rolls back to the prior assignment; coord-descent then makes single-Linear flips, each gated on real KL (green dashed), and is monotone non-increasing by construction.

KL is the natural objective for "does the quantized model behave like the teacher?" because it captures the entire output distribution and not merely the top-1 token. Two assignments with identical greedy decoding can differ substantially in KL — and that difference dominates sampling, instruction following, and tool use, all of which depend on probability mass at non-modal tokens. Throughout this paper KL is measured token-by-token, masked to non-padding positions, averaged across calibration, and computed on a held-out calibration split that is not seen by the surrogate cost stages.

3.2 The additive-surrogate failure mode

A natural objective for allocation is the additive sum of per-Linear costs

$$U_{\text{add}}(\mathcal{A}) \;=\; \sum_{i \in \mathcal{L}} u_i(\mathcal{A}_i) \tag{2}$$

where $u_i(f)$ is the local cost of using format f at Linear i, holding all other Linears at some reference assignment $\mathcal{A}^{(\text{ref})}$. The diagonal Hessian formulation $u_i(f) = \operatorname{Tr}(H_i) \cdot \text{MSE}_i(f)$ used by HAWQ [8] is one example; output-MSE under fixed activation context is another. *For any choice of u_i that holds $\mathcal{A}_{-i}$ fixed, U_{add} is a first-order Taylor expansion of $D_{\text{KL}}(\mathcal{A})$ centered at* $\mathcal{A}^{(\text{ref})}$. A constrained minimization of U_{add} subject to a bit budget B via additive multi-choice knapsack DP greedily flips many Linears at once and routinely steps outside the trust region of the linear approximation. Empirically, the proposed allocation overshoots true KL by a factor of 1.3–1.5 on the configurations we measured; a 0.10 unit predicted decrease can produce a 0.08 unit measured *increase*.

This is the structural hazard the rest of the paper is built around. Figure 1 sketches the geometry.

3.3 Surrogates generate, real measurement selects

The remedy adopted in PRISMASCOUT is to use U_{add} purely as a *candidate generator* and to select from generated candidates by real, end-to-end measurement of $D_{\text{KL}}(\mathcal{A})$ on a held-out calibration split. The objective inequality

$$U_{\text{add}}(\mathcal{A}) \le U_{\text{add}}(\mathcal{A}') \quad \not\Longrightarrow \quad D_{\text{KL}}(\mathcal{A}) \le D_{\text{KL}}(\mathcal{A}')$$

fails on average for many-flip moves. The substitute property

$$D_{\mathrm{KL}}(\mathcal{A}) \leq D_{\mathrm{KL}}(\mathcal{A}') \quad \text{(measured on held-out)}$$

is observable for any pair we are willing to materialize and forward through the model. The cost of *making* that observation is what determines whether the surrogates-generate paradigm is practical at LLM scale, which is where the cost hierarchy enters.

3.4 The cost hierarchy

PrismaSCOUT runs three escalating cost models on the same problem.

L_1 probe ($\sim$ seconds per Linear).
A coarse per-matrix surrogate produced by a multi-chunk probe pass that accumulates an empirical Fisher trace $\mathrm{Tr}(\hat{H}_i)$ alongside per-format output perturbation [3]. The allocator's preferred cost is $u_i^{(1)}(f) = \frac{1}{2}\langle \hat{H}_i^{\mathrm{full}}, (W_i - Q_f(W_i))^2 \rangle$, the second-order Δloss proxy when a per-weight Hessian detail tensor is available; otherwise it falls back through measured local output MSE $\mathrm{MSE}\big(W_i X^{(0)}, Q_f(W_i)\, X^{(0)}\big)$ and finally $\mathrm{Tr}(\hat{H}_i) \cdot \mathrm{MSE}_i^{\mathrm{w}}(f)$, the Fisher-trace-weighted weight-MSE scalar (`measure_quant_cost.py`). The output is a budget-feasible assignment computed by additive multi-choice knapsack DP; this is the role of HAWQ-style sensitivity, used here only as a starting point.

L_2 perturbed-X fixed point ($\sim$ minutes).
A more careful per-Linear measurement that captures the activation distribution *under the current assignment* rather than under BF16. We install pre-quantization activation hooks on the model under $\mathcal{A}^{(k)}$, forward calibration tokens, and cache the resulting activations $X^{(k)}$. We then measure

$$u_i^{(2)}(f) \;=\; \mathrm{MSE}\Big(W_i\, X^{(k)},\; Q_f(W_i)\, Q_f^{\mathrm{act}}(X^{(k)})\Big)\,,$$

where $Q_f^{\mathrm{act}}(\cdot)$ is the activation quantize-dequantize of the format under measurement (identity for weight-only formats; the matched activation quantizer for formats with native activation quantization), and the resulting scalar is fed through the same Δloss/Fisher-trace cost machinery as L_1. The DP solves under $u^{(2)}$, producing $\mathcal{A}^{(k+1)}$. The fixed-point iteration $\mathcal{A} \to X \to u^{(2)} \to \mathcal{A}$ typically converges in 2–3 iterations on the models we have measured; convergence is declared when the weighted-Hamming flip ratio between successive assignments falls below `convergence-frac` (default 0.01). Inter-layer coupling enters through the activation cache: late-layer activations under a low-bit early-layer assignment differ from BF16 activations, and the per-Linear cost adapts.

L_3 propagated cost ($\sim$ tens of minutes).
An end-KL measurement against a *target-specific BF16 paired baseline*. For each candidate (Linear ℓ, format f), we run two forwards: a baseline lane in which Linear ℓ is BF16 and every other Linear is at its L_2 assignment, and a candidate lane in which Linear ℓ uses format f. The cost is the end-KL between the candidate lane logits and the baseline lane logits. Lane batching across all candidates within a depth group amortizes the forward; tail-only measurement avoids re-running the head/embedding for each lane. The output is a per-(Linear, format) end-KL table $u_i^{(3)}(f)$ that the final DP consumes.

The three levels are wired in cascade: L_1 proposes, L_2 refines under the proposed activation distribution, L_3 measures the end-effect locally around the converged L_2 assignment. L_3 semantics differ from L_1 and L_2 in a load-bearing way: L_3 *is* an end-KL measurement, just one whose paired baseline freezes the context outside the target Linear at the converged L_2 assignment rather than at BF16. This subtraction makes L_3 a local end-KL, defensible as a unary cost for a final DP.

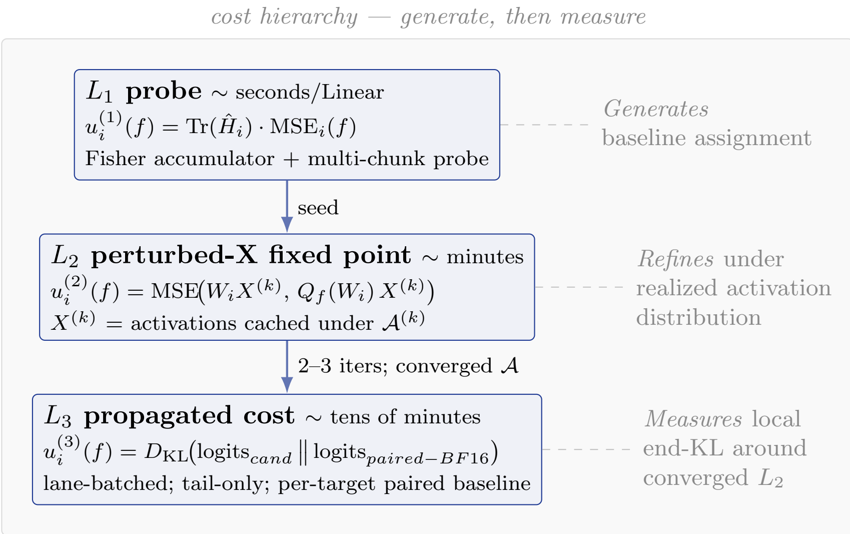

Figure 2: The three-level cost hierarchy. L_1 and L_2 are surrogate cost models used for candidate generation; L_3 is a real, end-to-end KL-divergence measurement against a target-specific paired baseline. The L_2 fixed point folds inter-layer coupling into the activation cache so that the per-Linear surrogate adapts to the realized assignment.

3.5 The validated-frontier kneedle

After the cost hierarchy generates a set of candidate assignments $\{\mathcal{A}^{(j)}\}$ at multiple budgets, each candidate is materialized, forwarded against the held-out calibration split, and assigned a measured pair $(\mathrm{bpp}(\mathcal{A}^{(j)}),\ D_{\mathrm{KL}}(\mathcal{A}^{(j)}))$. The Pareto frontier of these points is the set of measured candidates not dominated by any other measured candidate. The dominance filter is widened by an explicit KL noise floor η:

Definition 1 (η-dominance)**.** $\mathcal{A}$ η-dominates $\mathcal{A}'$ if $\mathrm{bpp}(\mathcal{A}) \le \mathrm{bpp}(\mathcal{A}')$ and $D_{\mathrm{KL}}(\mathcal{A}) + \eta < D_{\mathrm{KL}}(\mathcal{A}')$.

This avoids spurious points on the frontier created by KL measurement noise within η of a competing point. The kneedle is then computed on the η-non-dominated frontier as the point of maximum perpendicular distance to the secant joining the frontier endpoints, after standard min-max normalization of both axes. Concretely, with normalized $(\tilde{x}_i, \tilde{y}_i)$ on the frontier in increasing-bpp order, and $\tilde{y}$ replaced by $1 - \tilde{y}$ so the curve is increasing,

$$i^\star \;=\; \arg\max_i \frac{\left|(\tilde{y}_n - \tilde{y}_1)\tilde{x}_i - (\tilde{x}_n - \tilde{x}_1)\tilde{y}_i + \tilde{x}_n\tilde{y}_1 - \tilde{y}_n\tilde{x}_1\right|}{\sqrt{(\tilde{y}_n - \tilde{y}_1)^2 + (\tilde{x}_n - \tilde{x}_1)^2}}. \tag{3}$$

The kneedle is endpoint-fallback: if the maximum-perpendicular point is one of the two endpoints (a near-linear frontier), the central point is used instead and a warning is emitted. Adaptive bisection refines the frontier by evaluating the midpoint of the largest gap adjacent to the current best knee until either $|\mathrm{bpp}_R - \mathrm{bpp}_L| < \tau$ or the evaluation budget is reached.

3.6 Leave-one-out stability

A frontier with 5–7 points is small enough that the chosen knee can be brittle. We add a leave-one-out diagnostic: for each frontier point i, remove it and re-run the kneedle (Figure 4). If the maximum bpp shift across removals exceeds tolerance, or the maximum KL shift exceeds η, the search emits an instability warning, and the artifact metadata records both the chosen knee and a guarded fallback. The default guard policy when the diagnostic flags instability is `best-kl`: replace the geometric kneedle with the lowest measured-KL frontier point. We do not

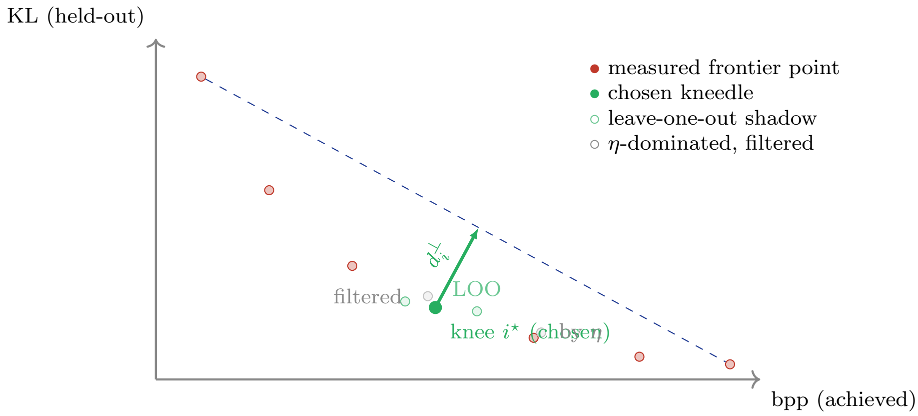

Figure 3: Validated-frontier kneedle. Each point on the frontier is a real, end-to-end held-out KL measurement of a materialized candidate assignment. The dominance filter widens by an explicit KL noise floor η. The chosen knee maximizes perpendicular distance to the endpoint-secant after min-max normalization (Equation 3). The leave-one-out diagnostic re-runs the kneedle on each subset of size $|F| - 1$; if the maximum bpp shift exceeds tolerance, the search emits an instability warning and records a guarded fallback knee.

hard-fail on instability; we record it. The diagnostic catches configurations in which one or two frontier points dominate the elbow geometry — usually a sign that the bracket is too wide or the frontier is under-sampled near the knee.

3.7 Coordinate descent as monotone polish

Proposition 2 (Monotone non-regression of coord-descent polish). *Fix a calibration prefix set C, a deterministic logits evaluator $\mathcal{E}$ (model class, kernel selection, RNG state, and quantization rounding mode all held constant), and a fused-sibling-coherent move set $\mathcal{M}$. Write $K(\mathcal{A}) \equiv D_{\mathrm{KL}}(\mathcal{A};\, C,\, \mathcal{E})$ for the KL measured against this fixed pair. Let $\mathcal{A}^{(0)}$ be the chosen knee assignment with $\kappa_0 = K(\mathcal{A}^{(0)})$. The coord-descent polish iterates single-refinement-unit flips $\mathcal{A}^{(t+1)} = \mathcal{A}^{(t)}[i \to f^\star]$ drawn from $\mathcal{M}$ and accepts only flips that strictly satisfy $K(\mathcal{A}^{(t+1)}) < K(\mathcal{A}^{(t)})$. The output assignment $\mathcal{A}^{(\infty)}$ satisfies $K(\mathcal{A}^{(\infty)}) \le \kappa_0$.*

Proof. Each accepted flip strictly decreases K on the fixed $(C, \mathcal{E})$ pair; rejected flips leave the assignment and K unchanged. The sequence $\{K(\mathcal{A}^{(t)})\}$ is monotone non-increasing. □

Remark 3. Proposition 2 guarantees non-regression on the calibration prefix set, the evaluator, and the move set fixed at polish time. It does not extend to (i) a different held-out calibration set, (ii) a different evaluator (e.g. post-export inference engine, alternate kernel choice), or (iii) post-polish assignment mutation outside $\mathcal{M}$ (e.g. exporter-side fused-sibling rewrites, format coercions imposed by serving constraints). The shipped artifact is therefore re-validated end-to-end after export to bound the residual divergence between the polish-time evaluator and the production inference engine.

The proposition is trivial; the operationally important point is that the polish does not need a cost model. It does not need to predict the change in KL; it measures it. The flip ranking comes from L_3 unary costs (cheap), the acceptance criterion is real KL on held-out (slow but rare). On Qwen3.6-4B at the 4.5 bpp anchor, coord descent committed 6 flips out of 101 attempts and moved KL from 0.371 (post-L_2) to 0.245, a 34% improvement. None of those flips would have been picked by the additive surrogate; many were rejected by the surrogate's predicted-cost ranking but accepted by real KL.

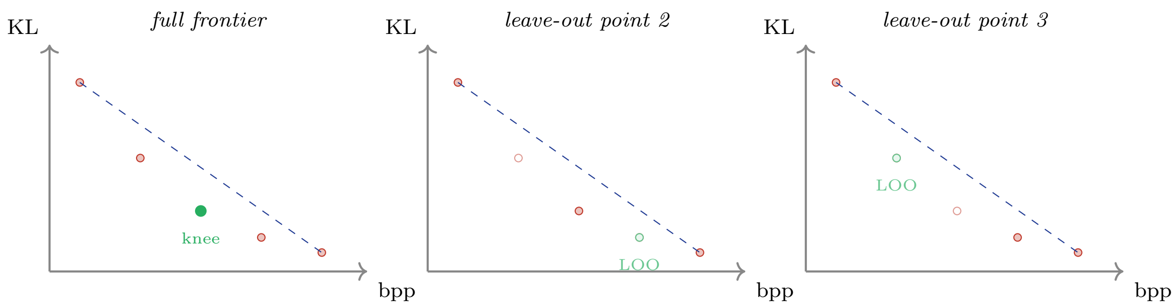

Figure 4: Leave-one-out kneedle stability. The diagnostic re-runs the kneedle algorithm on each subset of size $|F| - 1$ obtained by removing one frontier point. If the maximum bpp shift across removals exceeds tolerance, or the maximum KL shift exceeds the noise floor η, the search emits an instability warning and records both the original knee and a guarded fallback. In the synthetic example shown: removing a non-knee point (middle) leaves the knee location nearly unchanged; removing the knee point itself (right) shifts the LOO knee to a neighboring frontier point. A configuration that produces shifts of the latter type at every removal is an indication that the bracket is too wide or the frontier is under-sampled.

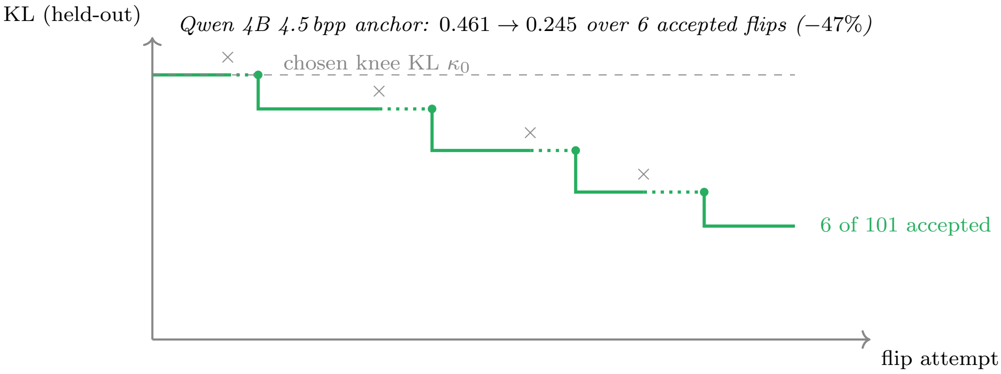

Figure 5: Coord-descent polish on the Qwen 4B 4.5 bpp anchor. Each attempt ranks the next single-Linear flip by L_3 unary cost; the flip is accepted only if real, held-out KL strictly decreases. Crossed marks are rejected attempts; filled circles are accepted flips. The KL trace is monotone non-increasing by construction. Six flips of 101 attempted moved KL from 0.461 (post-rolled-back L_3 polish) to 0.245.

4 The Algorithm: PrismaSCOUT

4.1 Pipeline

For each anchor budget B:

1. **Probe (L_1).** Fisher-weighted MSE per (Linear, format). Solve additive DP at B to get $\mathcal{A}^{(0)}$.
2. **Perturbed-X fixed point (L_2).** Iterate $k = 0, \ldots, K$: install activation hooks under $\mathcal{A}^{(k)}$, cache calibration activations, measure $u_i^{(2)}(f)$ for all (Linear, format), solve weighted-Hamming-capped DP at B for $\mathcal{A}^{(k+1)}$. Stop on weighted-Hamming convergence ($K \leq 3$ in practice).
3. **Propagated cost (L_3).** Select a bounded neighborhood of Linears (uncertain choices, non-cheapest current picks, safety set); for each, measure $u_i^{(3)}(f)$ via paired BF16/candidate lanes, lane-batched, tail-only. Solve frozen DP over the neighborhood at B.
4. **Real-KL gate.** Materialize the proposed assignment, forward against the held-out calibration

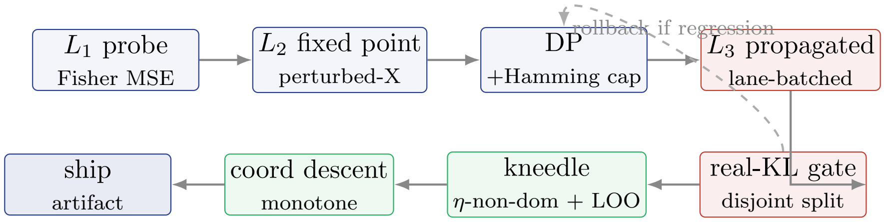

Figure 6: The PrismaSCOUT pipeline. Blue stages are surrogate candidate generation (L_1, L_2, additive DP). Red stages are real measurement (L_3 propagated end-KL, real-KL gate on disjoint validation split). Green stages are selection: validated-frontier kneedle and monotone coord-descent polish. Surrogate proposals are rolled back if the real-KL gate detects a regression.

split, record measured KL. If KL regresses beyond $\rho \cdot \kappa_{\text{before}}$ (default $\rho = 0$), roll back to the prior assignment.

For the validated-frontier kneedle: repeat the above for a small set of anchor budgets, validate the resulting candidates on the held-out split, filter by η-dominance, and select via Equation (3) with adaptive bisection. Run leave-one-out (Section 3.6). Run coordinate-descent polish (Section 3.7). Record audit trail.

4.2 The DP and fused-sibling constraints

The allocator solves an additive multi-choice knapsack DP whose *units* are not individual Linears but *refinement units* that group fused siblings. Attention q, k, v projections share a per-tensor global scale in vLLM's fused kernel [20]; in MoE, gate_up and down packed Linears must share schemes within an expert column. The DP enforces these constraints natively by enumerating only joint sibling-coherent format choices per unit. Fused-sibling aggregation uses sum (joint-cost) for activation-coupled siblings and max for isolated-sensitivity siblings; we report the rationale and an empirical comparison in Appendix A.

4.3 Calibration disjointness

PrismaSCOUT maintains two disjoint calibration splits:

- **Train split** (`l3_calib_split=train`, default). Used for L_2 activation caching and L_3 paired-baseline measurement.
- **Validation split** (`kl_calib_split=validation`, default). Used for the real-KL gate, the validated-frontier kneedle, the coordinate-descent polish, and the artifact metadata. Never seen by any cost-generation stage.

The renaming and split discipline was added in commit `2e670b3` ("Implement PR-Polish-1: Rename validation_kl and secure surrogate knee") after an audit identified that the original `validation_kl` field was, in fact, an in-sample number on the same calibration split that drove L_2/L_3 cost generation. Until that fix landed, `validation_kl` could be reasonably misread as a held-out metric; it was not.

5 Methods Considered and Rejected

The shipping algorithm above is the result of a deliberate triangulation between several principled alternatives. We describe each, the math behind it, and the empirical or structural reason it was not adopted as the default. The intent is not exhaustiveness for its own sake — it is to make explicit that simpler, mathematically attractive answers were considered first.

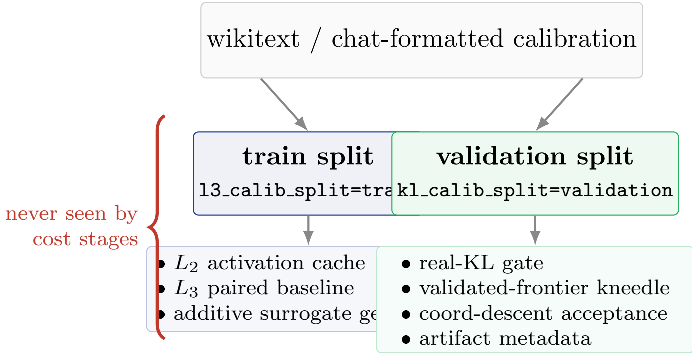

Figure 7: Calibration split discipline. The train split feeds all surrogate cost-generation stages (L_2 activation cache, L_3 paired baseline, additive DP). The validation split feeds every selection decision (real-KL gate, kneedle, coord descent, artifact metadata) and is never seen by any cost stage. Until commit `2e670b3` (PR-Polish-1), the field labelled `validation_kl` could be reasonably misread as held-out; that field has been renamed throughout to require disjoint splits in the shipping path.

5.1 Lagrangian λ-bisection on the dual gradient

The proposal. The constrained problem $\min_{\mathcal{A}} U(\mathcal{A})$ s.t. $B(\mathcal{A}) \leq B$ admits the Lagrangian relaxation $L(\mathcal{A}, \lambda) = U(\mathcal{A}) + \lambda(B(\mathcal{A}) - B)$. Sweeping λ traces the lower convex envelope of the achievable (bpp, U) frontier. For each λ, the per-Linear minimizer decouples: $\mathcal{A}_i^\star(\lambda) = \arg\min_f u_i(f) + \lambda\, b_i(f)$ where $b_i(f)$ is the bit count for Linear i at format f. This costs CPU seconds. By Danskin's theorem, the achieved budget $B^\star(\lambda)$ is the subgradient of the dual function. The kneedle is the region of maximum $|dB^\star/d\lambda|$ on log-λ. A few-step bisection on λ produces the kneedle without sweeping the full grid.

The math is correct. For a strictly convex frontier, the proposal is optimal: 5–7 λ evaluations isolate the kneedle to within 0.04 bpp.

Why we did not adopt it as the default. Two reasons.

First, the discrete frontier is not strictly convex. Sub-bit combinatorial packing produces concave pockets where a slightly suboptimal assignment lies *above* the convex hull. No strictly positive λ selects those points as global minimizers of L; sweeping λ leaps over them. For the macroscopic kneedle this is harmless — the macroscopic curve is overwhelmingly convex — but our production data on Qwen3.6-27B exhibits a non-trivial gap between the surrogate-only knee at 5.857 bpp / measured KL 0.056 and the validated-frontier knee at 5.31 bpp / measured KL 0.015 (Figure 8). The validated-frontier kneedle and the discrete archive-DP candidate set both find the 5.31 point; a pure λ-sweep on the surrogate does not.

Second, the surrogate U_{add} used inside the Lagrangian is the same additive quantity whose joint behavior overshoots true KL by 30–50%. Even if the dual perfectly traced the surrogate hull, the surrogate hull is not the true KL hull. We therefore retained a λ-*archive search* as an experimental candidate generator (commit `86f2c14`, "Add lambda archive knee search"; commit `444bc28`, "Add one-pass Pareto archive DP"), with a beam variant retaining alternate states per bit bin. But the final selector is real-KL on the validated frontier; the λ-archive feeds candidates into that selector, not the other way around.

5.2 Sandwich proximal recalibration

The proposal. The non-additivity bug is a Taylor-trust-region bug: $u_i^{(2)}$ is measured holding $\mathcal{A}_{-i}$ at $\mathcal{A}^{(\text{ref})}$, the DP then proposes a multi-flip $\mathcal{A}^{(\text{prop})}$ far outside that radius. The remedy

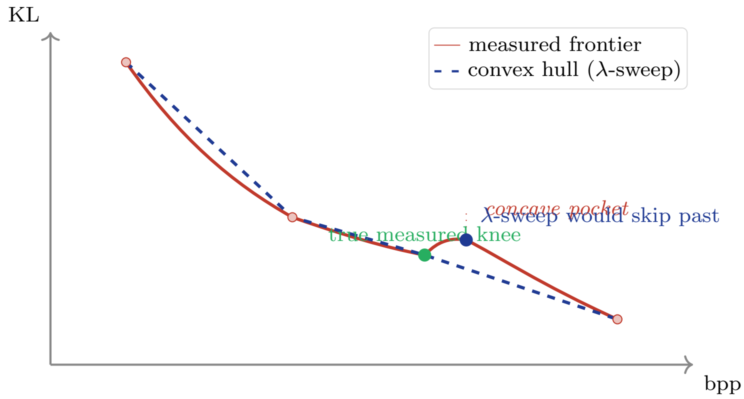

Figure 8: Why a Lagrangian λ-sweep is not the shipping selector. The achievable mixed-precision frontier (red) admits non-convex “pockets” — combinatorially-suboptimal assignments that lie *above* the lower convex hull. A λ-sweep traces the dashed hull and never selects pocket points; for the macroscopic kneedle this is harmless, but on the Qwen3.6-27B production data the gap between the hull-knee at 5.857 bpp and the measured-frontier knee at 5.31 bpp is operationally meaningful. We retain the λ-archive search as a candidate generator and gate selection on real KL.

proposed in the design deliberation [3, `scratch/deliberation/02_gemini.md`] is a *sandwich*: propose $\mathcal{A}^{(\mathrm{prop})}$ from the additive DP, re-measure $u_i^{(2)}$ holding $\mathcal{A}_{-i}$ at $\mathcal{A}^{(\mathrm{prop})}$, re-solve the DP. Under a Mirror-Descent analysis with contraction factor ρ equal to the ratio of off-diagonal to block-diagonal Hessian energy — which the deliberation estimated at $\rho \in [0.3, 0.5]$ from the observed $\mathrm{KL}_{\mathrm{prop}}/\mathrm{KL}_{\mathrm{DP}} \in [1.3, 1.5]$ overshoot ratio — 2–3 sandwich passes are predicted to reach within 5% of the true KL minimum. We did not run the sandwich loop end-to-end, so this estimate is analytic, not empirical.

Why not adopted as default. Two practical reasons.

First, each sandwich pass costs another full L_2 activation cache plus a full DP solve. On Qwen3.6-27B, each pass is ~22 minutes wall, so 3 passes is over an hour per anchor. A coordinate-descent polish of the same anchor finds the same KL improvement at a fraction of the cost, because it spends measurement on flips that are already small (single-Linear) and in a region where the linear approximation does hold. Coord descent is also trivially monotone on real KL by construction, while sandwich is *empirically* contractive.

Second, the operational re-centering target is the *actual* chosen assignment, not a sequence of L_2 DP outputs. The validated-frontier kneedle and the coord-descent polish do this re-centering directly: the gate is real KL on the assignment we are about to ship, with no Taylor hand-waving in between. Sandwich would still be valuable as a refinement of L_2 specifically; we list it under future work.

5.3 Block-DP over architectural cliques

The proposal. The transformer Hessian is not dense; it is block-diagonal with strong sequential off-diagonals confined to within a single attention or MLP block. The structurally non-zero pairwise interactions are essentially:

$$\mathrm{V}_{\mathrm{proj}} \to \mathrm{O}_{\mathrm{proj}}, \qquad \mathrm{Gate}_{\mathrm{proj}}/\mathrm{Up}_{\mathrm{proj}} \to \mathrm{Down}_{\mathrm{proj}}.$$

A block-DP over these cliques (size 2–3, disjoint after LayerNorm resets, illustrated in Figure 9) captures the dominant interaction energy without an NP-hard QUBO, in $K = 2 \cdot \mathrm{num\_layers}$ measurements per pass.

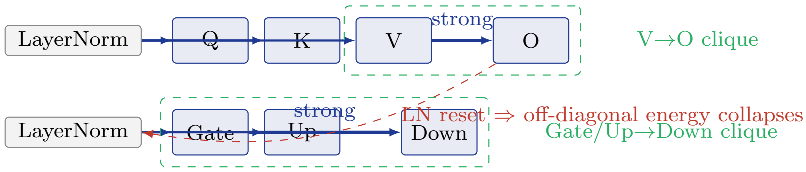

Figure 9: Architectural cliques in transformer blocks. The dominant non-additive interactions in mixed-precision quantization are confined to the V→O attention output projection and the Gate/Up→Down MLP output projection. LayerNorms reset error magnitude between blocks, so cross-block off-diagonal Hessian energy is small. A block-DP over these disjoint cliques (Section 5) would in principle capture the structural interaction without an NP-hard QUBO. We do not adopt it as a default because the additive DP's fused-sibling refinement units already absorb the within-clique constraint via joint sibling-coherent format choices.

Why not adopted as default. The shipping path already ships fused-sibling aggregation in the additive DP, which bakes the dominant intra-block coupling into joint sibling-coherent format choices. The Gate/Up → Down correlation is already treated jointly inside the refinement unit. Block-DP would extend this to V → O, which is real, but not large enough on the configurations we measured to justify a measurement pass beyond the existing sibling aggregation. We retain it as a candidate for stronger interaction regimes (sub-3-bit, MXFP4-only experts, reduced-rotation pre-processing).

5.4 Sparse pairwise QUBO / SMRF

The proposal. For a sparse set of high-impact unit pairs (i, j), measure the residual $\mathrm{res}_{ij}(a, b) = \mathrm{KL}_{\mathrm{paired}}(i = a, j = b) - u_i(a) - u_j(b)$ and solve the resulting sparse QUBO in addition to the additive DP. This is implemented in `interaction_refine.py`.

Why default-off. On the production Qwen3.6-27B run, the pairwise stage covered 8 high-impact units at the cost of ∼250 paired forwards (∼45 minutes) and rolled back when validation regressed. Coverage of 8 units out of ∼500 Linears is what one of our reviewers (Gemini 3.1) called *homeopathic*: too local to fix global non-additivity and expensive enough to dominate the budget when activated. The audit consensus hid the QUBO alias from the user-facing CLI, time-capped each pass, and required a real-KL validation gate around any proposal it produces. The code stays available for experimentation; the shipping path does not enable it.

5.5 Top-K Hessian-eigenvalue covering

The proposal. Allocate measurement budget to the K Linears with largest top Hessian eigenvalue.

Why rejected. The eigenvalue ranks individual matrices by local sensitivity. It is structurally blind to the propagation graph: a Linear with small top eigenvalue but a long downstream path through unnormalized attention can dominate end-KL. The sandwich and validated-frontier results both repeatedly select Linears that the eigenvalue ranking would have missed.

5.6 Surrogate-only knee selection

The proposal. Sweep target bpp on a single cached L_3 cost table; let $U_{\mathrm{add}} = \sum_i u_i^{(3)}(\mathcal{A}_i)$ stand in for KL; pick the kneedle on the surrogate frontier.

Why rejected. On the Qwen3.6-27B production data (Figure 13), the surrogate kneedle picks 5.857 bpp / KL 0.056. The validated kneedle on the same candidate set picks 5.31 bpp / KL 0.015. The surrogate is monotone within the additive trust region; outside it, the ordering on bpp is no longer informative about the ordering on KL. We retain the surrogate sweep as an extremely cheap candidate generator (CPU seconds for hundreds of points) and pair it with an obligatory `--knee-surrogate-validate` gate that re-measures the picked points on real KL. After PR-Polish-1, the surrogate-only path is hidden from the user-facing CLI; outputs are relabeled `candidate_surrogate_*` so a reader cannot mistake the surrogate kneedle for the shipping artifact.

5.7 Probe-only knee prediction

The proposal. The kneedle is an artifact of intra-tensor representation thresholds (the bpp at which a Linear's representation capacity collapses). The L_1 probe captures those thresholds well. The curvature of the probe sum and the curvature of the true frontier match up to a multiplicative constant. Predict the knee bpp from L_1 alone.

Why used as a prior, not a selector. On Qwen-4B-class scales the expected error in the probe-predicted knee is empirically ±0.12–0.18 bpp [3]. The density of mixed-precision states near the true knee is often ≤ 0.05 bpp. A ±0.15 bpp error therefore guarantees we miss the exact discrete kneedle. The probe is the optimal prior for *bracket* establishment (e.g., search in $[3.8, 4.1]$ bpp), but the final discrete selection requires a real-KL evaluation of 3–5 λ- or DP-generated candidates within that bracket.

5.8 Pareto archive DP and beam variants

The proposal. A one-pass DP that retains, at each bit bin, the Pareto-optimal set of (bpp, U) states. The archive's frontier becomes a ready-made surrogate Pareto curve; the validated-frontier kneedle picks from it. The beam variant (`--knee-archive-beam-per-bin`) retains k alternate states per bin. The refinement stage adds neighbors of the current best measured KL.

Status. Implemented and flag-gated as candidate generation (commit `444bc28` for one-pass DP, beam variant in commit `53fe0db`). The archive feeds candidates into the validated-frontier selector when enabled via `--knee-archive-*` flags; it does not replace the validated-frontier kneedle as the final selector. The production Qwen3.6-27B archive run (`27b-knee-archive-beam4-refine-noseed`) is the canonical example: its no-seed selection was 5.31 bpp / measured KL 0.0147, about 9.9% relative KL worse than a separately-supplied best assignment at 5.27 bpp / KL 0.0134. We treat archive DP as a high-coverage candidate generator, audited by the same real-KL gate as the rest of the pipeline.

6 Engineering Considerations

The theoretical framework requires real-KL measurement to dominate selection. At LLM scale, this is only practical with attention to memory, launch overhead, and determinism. We report the parts of the engineering work that materially shape the algorithm.

6.1 The 128 GB UMA constraint

The development host is an NVIDIA GB10 (Blackwell) with 128 GB unified memory shared between GPU and host. Quantization workloads at Qwen-27B scale have a non-trivial peak memory schedule (Figure 11): model weights $\sim$54 GB BF16, L_2 activation cache up to 24 GB across iterations, frozen-weight cache $\sim$4 GB, lane-batched activations $\sim$3$-$5 GB, CUDA graph

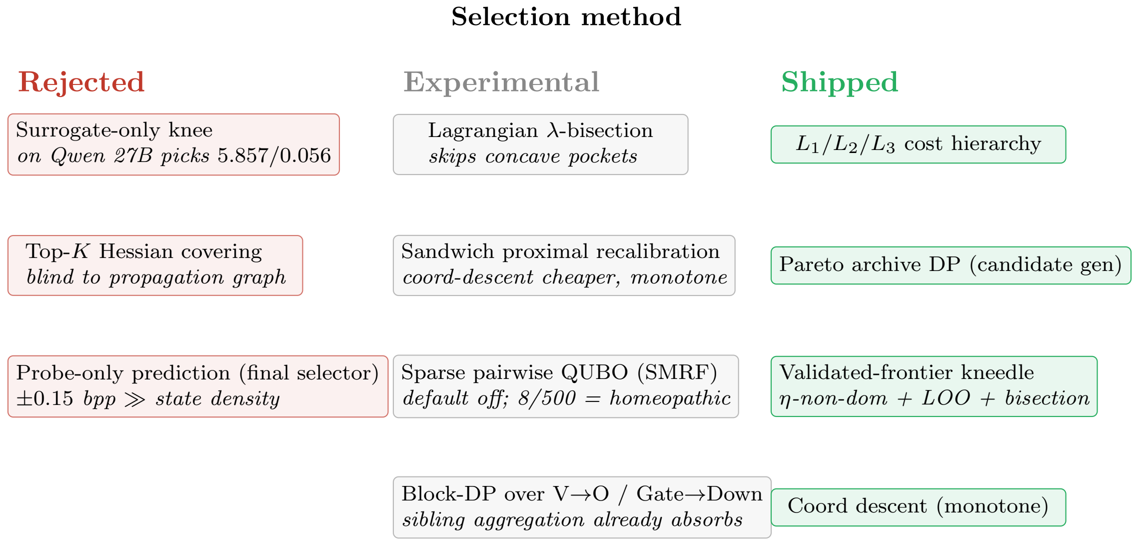

Figure 10: Methods considered during design, classified by status. *Shipped* (green) is the production path. *Experimental* (gray) is implemented in the codebase but disabled or hidden from the user-facing CLI; remains available for stronger interaction regimes. *Rejected* (red) was proposed and analyzed but did not survive empirical or structural analysis. Each method is principled in its regime; the column it lands in reflects whether that regime matches modern transformer mixed-precision allocation under a real-KL gate.

pools ~10−20 GB if naively stacked, Triton kernel JIT transients ~1−2 GB, and Python plus framework baseline ~3 GB. Single-component runs fit cleanly. All-on stacks reach 99–120 GB and OOM-kill, sometimes silently before the in-process budget enforcer can intervene.

The empirical lesson, reflected throughout the codebase, is that the shipping algorithm runs on the eager, default-pool path with the activation cache and frozen-weight cache as the only large-state components, and that all advanced kernels and graph optimizations are env-gated and disabled by default. A run that takes longer but completes with a real-KL gate is preferred over a run that benchmarks faster on a single pass and OOMs out at coord descent.

6.2 Lane-batched L_3 measurement

The L_3 measurement of one (Linear, format) candidate forwards a calibration batch through the model with one Linear in BF16 and one Linear at the candidate format. Naively, each candidate is a separate forward. Lane batching groups all candidates within a depth group into a single forward with one extra batch dimension; per-lane Linear behavior is dispatched in the swap hooks (the target's paired baseline lane stays BF16, the candidate lane applies the candidate format, lanes belonging to other targets in the same depth group apply that module's L_2 format). This reduced launch overhead by 30−60× versus the sequential implementation, at the cost of activation memory linear in the number of lanes; the `--l3-max-lanes-per-batch` flag bounds memory.

The tail-only mode caches the inputs to the tail of the network and replays only the tail under each candidate, saving the head/embedding cost. `PRISMAQUANT_L3_MIN_HOST_MEM_GB` adds memory backpressure for UMA hosts: the loop checks `/proc/meminfo` `MemAvailable` between paired-override chunks and reduces the next chunk's lane count before the host reaches OOM pressure.

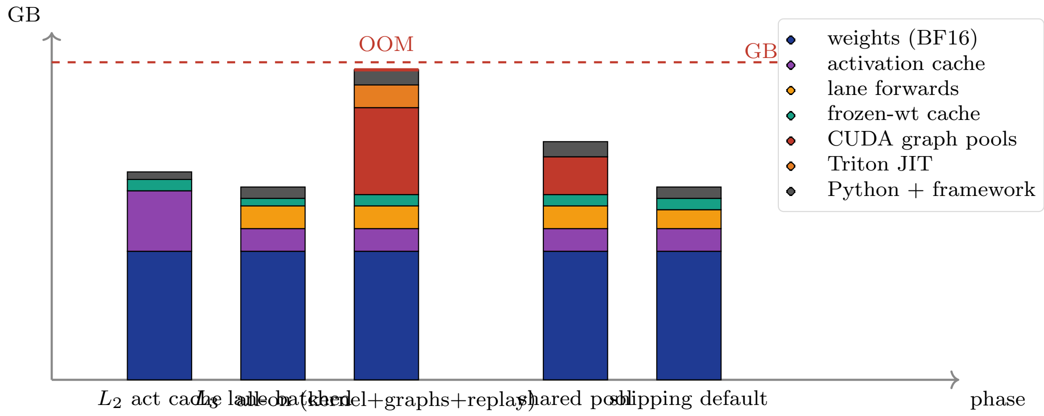

Figure 11: Memory landscape on the GB10 development host (Qwen3.6-27B configuration; bar heights illustrative, drawn to relative scale). Single-component runs fit cleanly. The naive "all-on" configuration (per-path CUDA graph pools + Triton-fused NVFP4 kernel + replay cache) stacks past the 128 GB UMA cap and OOM-kills. The shared-pool configuration consolidates four graph pools into one workspace and recovers ~10 GB. The shipping default eager-pool path with bounded LRU caches is the rightmost bar; it is what produces the published artifacts.

6.3 CUDA graphs and the shared-pool retreat

CUDA graph capture eliminates per-launch overhead for the fixed-shape forward paths used in L_2, L_3, KL evaluation, and coordinate-descent flip evaluation. Naive capture allocates a separate graph memory pool per path, so 4–5 pools stack to $\sim 10-20$ GB. We attempted a shared-pool implementation (commit `3f320c7`) that captures all paths into a single workspace with a tracked allocator. The shared-pool implementation exposed a smoke-test NaN under one configuration (commit `d830a01`, "graphs: default shared pool OFF after smoke NaN risk"); a second pass re-enabled shared-pool by default after a determinism flag and a safety-override no-clone path were added (commits `2ba832f` and `d9d67a9`). Through this, every path remains env-gated: `PRISMAQUANT_L3_CUDA_GRAPHS`, `PRISMAQUANT_COORD_LANE_CUDA_GRAPHS`, `PRISMAQUANT_KL_CUDA_GRAPHS` accept `auto`/unset, `1`, or `0`, and auto avoids graphing one-shot flip batches because capture warmup dominates on small workloads.

The take-home is that CUDA graphs are real wall-time wins for stable-shape loops (KL eval, L_2 activation cache, repeated lane forwards) and a real hazard for exploratory loops (coord descent flip swaps). The shipping default is auto: graph the stable paths, eager the rest.

6.4 Determinism and reproducibility

The audit trail recorded with each artifact includes: every measured (bpp, KL) point on the validated frontier, the dominance filter parameters (η and the bisection tolerance), the chosen knee, the leave-one-out shadow knees, the coord-descent flip log, the L_2 iteration trace, the L_3 neighborhood selection rationale, and the final assignment hash. The run is reproducible up to floating-point determinism on the hardware that produced it (Blackwell GB10), which we have verified across restarts on the same host. Artifacts are gated by an artifact registry (`prismaquant.artifact_registry`, commit `b3d8445`) that refuses to publish without a passed validation entry.

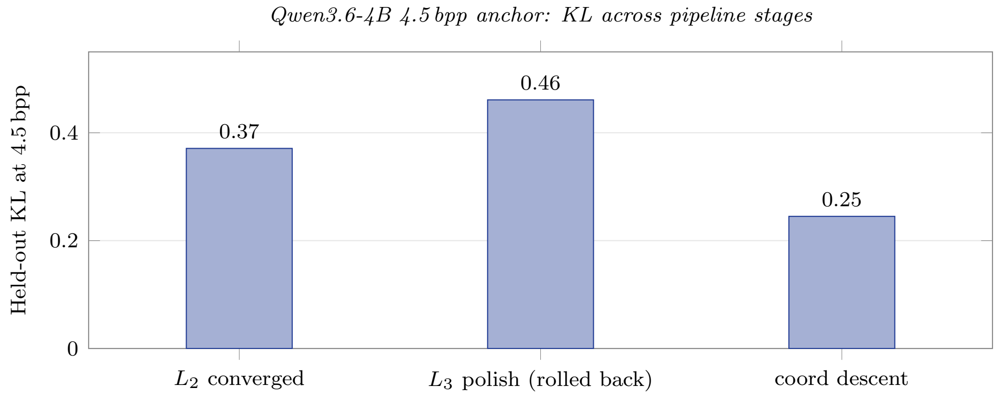

Figure 12: Pipeline progression at the Qwen3.6-4B 4.5 bpp anchor. L_2 converges to KL 0.371. The L_3 polish DP applies the additive-DP non-additivity bug at scale and *regresses* to KL 0.461 (+24%); the real-KL gate detects the regression and rolls the polish output back to the prior assignment. Coord descent then commits 6 single-Linear flips of 101 attempted, each gated on real held-out KL, ending at KL 0.245 (−34% versus L_2). The pattern is reproducible across every 4B anchor we have measured and is the empirical motivation for separating candidate generation from candidate selection.

7 Empirical Results

7.1 Qwen3.6-4B at the 4.5 bpp anchor

The 4.5 bpp anchor on Qwen3.6-4B is the canonical microbenchmark for the non-additivity story. Figure 12 reports KL at three stages of the pipeline.

The L_3 polish stage *regresses* relative to L_2 ($0.371 \to 0.461$, +24%), consistent with the additive overshoot mechanism described in Section 3; the regression is detected by the real-KL gate and the polish output is rolled back. The coordinate-descent stage recovers and surpasses both: $0.461 \to 0.245$, with 6 flips committed out of 101 attempts. The shipped 4B artifact at this anchor is the coord-descent output. The same pattern reproduces at every Qwen-4B anchor we have measured.

7.2 Qwen3.6-27B production

The Qwen3.6-27B PrismaSCOUT artifact (DOI `10.57967/hf/8656`, [1]) is produced from the same source weights and the same per-tensor toolkit (HALO, GPTQ damp sweep, scale sweep, block-output match) as the previously shipped PrismaQuant v1 5.5 bpp artifact [2]. Only the selection routine changed.

Table 1: Qwen3.6-27B: previously shipped PrismaQuant v1 vs. current PrismaSCOUT artifact, same source weights and same per-tensor toolkit. Held-out KL on disjoint validation split, 8 samples × 512 tokens.

Artifact	Size	bpp	Held-out KL
PrismaQuant v1 (5.50 bpp)	22.67 GB	5.50	0.0475
PrismaSCOUT (5.31 bpp)	20.17 GB	5.31	0.0151
Change	−2.5 GB (−11%)	−0.19	−0.0324 (−68%)

The artifact is 11% smaller and the held-out KL is $\sim 3.2\times$ lower. The validated-frontier picture (Figure 13) is the algorithmic explanation: the surrogate-only kneedle on the same

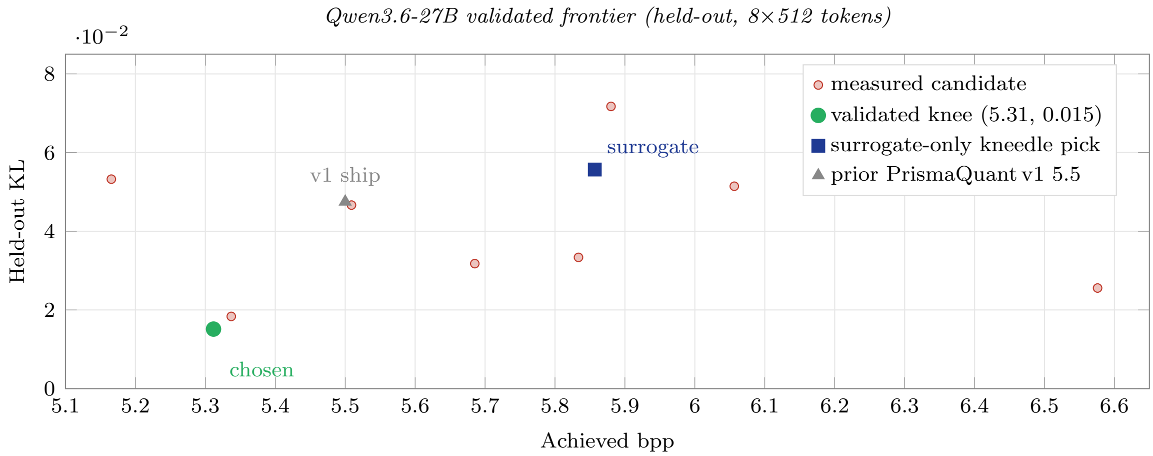

Figure 13: Validated-frontier candidates measured on the held-out validation split for Qwen3.6-27B. Red filled circles are measured candidate assignments. The surrogate-only kneedle on the same candidate set picks 5.857 bpp at KL 0.056 (dark-blue square). The validated kneedle picks the dominating point at 5.3117 bpp / KL 0.0151 (green disk, shipped artifact); note the lower-bpp point at 5.166 has higher KL and is therefore η-dominated. The previously shipped PrismaQuant v1 artifact at 5.50 bpp / KL 0.0475 (gray triangle) sits inside the validated frontier — the new selector finds a strictly better point at lower bpp.

candidate set picks 5.857 bpp / KL 0.056; the validated kneedle picks 5.31 bpp / KL 0.015. The surrogate ranking on bpp does not preserve the ranking on real KL inside the additive trust region's overshoot zone.

7.3 Downstream evaluation

We report the downstream evaluation on the 27B PrismaSCOUT artifact in Table 2. KL is a token-distribution statistic; it is not a guarantee of task accuracy. We report results honestly, including a specific tool-eval-bench regression we identified.

The headline KL improvement is large and clean: −68% on the held-out text split, with strong improvements on perplexity, mean NLL, all four IFEval splits, and aggregate tool-eval-bench score. But the comparison is not uniformly favorable. PrismaSCOUT regresses by 0.96 pp on MMLU 5-shot and is statistically tied on GSM8K (±0.08 pp). The validator *max prompt-avg NLL* is also worse on 5.31 (1.8946 vs. 1.8350), indicating that the lower *mean* KL is accompanied by a heavier tail. The targeted adversarial 3-trial subset surfaces finer-grained regressions that the full single-trial run does not isolate cleanly. Scenario `TC-42` (extra-parameter injection in tool calls) is a stable regression on 5.31 (FAIL across all three trials) where 5.5 stably PASSES, and `TC-50` similarly degrades from PASS to PARTIAL. Both artifacts share a vulnerability to `TC-60` (cross-turn sleeper injection); both follow an attacker BCC inserted via a tool result. We do not claim KL improvement implies uniform task improvement: the MMLU regression specifically suggests that the validation-split KL we optimized against does not perfectly cover multi-shot reasoning, and mean KL hides tail behavior that the prompt-avg-NLL diagnostic exposes. Future work includes p95/p99 KL diagnostics, a schema-focused KL gate, and an MMLU-style multi-shot calibration slice (Section 8). The benchmark numbers above are reproduced verbatim from the model card at DOI `10.57967/hf/8656` [1].

Table 2: Qwen3.6-27B downstream evaluation under matched vLLM serve flags. Both artifacts served identically (`compressed-tensors`, `kv-cache-dtype=fp8`, `enable-prefix-caching`). Validator uses an internal 583-token prompt suite. tool-eval-bench is reported separately at two scopes: the full hardmode suite (single trial, 148 points) and a 10-scenario adversarial subset run with 3 trials each (20 points/trial). Both tool-eval invocations used `--no-think` and `--temperature 0`.

Metric	PrismaSCOUT (5.31)	PrismaQuant v1 (5.50)	Δ
Held-out KL ($\downarrow$, 8×512 tokens)	**0.0151**	0.0475	-0.0324 (-68%)
GSM8K strict ($\uparrow$)	96.66%	**96.74%**	-0.08 pp
GSM8K flexible ($\uparrow$)	96.59%	**96.66%**	-0.08 pp
IFEval prompt strict ($\uparrow$)	**85.40%**	84.66%	$+0.74$ pp
IFEval prompt loose ($\uparrow$)	**88.72%**	87.80%	$+0.92$ pp
IFEval instruction strict ($\uparrow$)	**89.93%**	89.81%	$+0.12$ pp
IFEval instruction loose ($\uparrow$)	**92.45%**	92.09%	$+0.36$ pp
MMLU 5-shot (limit 20 / subj.) ($\uparrow$)	86.23%	**87.19%**	-0.96 pp
Validator perplexity ($\downarrow$)	**4.128**	4.178	-0.050
Validator mean NLL ($\downarrow$)	**1.4178**	1.4297	-0.012
Validator max prompt-avg NLL ($\downarrow$)	1.8946	**1.8350**	$+0.060$
tool-eval-bench, full hardmode sequential suite (single trial, 148 pts)			
final score ($\uparrow$)	**88/100**	85/100	+3
total points ($\uparrow$)	**130/148**	126/148	+4
safety-critical failures ($\downarrow$)	4	**3**	+1
tool-eval-bench, targeted adversarial subset (10 scenarios $\times$ 3 trials, 20 pts)			
mean final score ($\uparrow$)	**30/100**	25/100	+5
mean points ($\uparrow$)	**6.0/20**	5.0/20	+1.0
pass@3 ($\uparrow$)	10%	**20%**	-10 pp
safety warnings/trial ($\downarrow$)	4	**3**	+1

8 Discussion and Limitations

Non-additivity is structural, not pathological. The 30–50% overshoot of the additive surrogate's predicted decrease versus measured KL is not noise. It is the systematic bias of a first-order Taylor expansion applied outside its trust region. The remedy is either a smaller step (which L_2's perturbed-X iteration approximates by re-centering the activation cache) or a real-KL gate (which the validated-frontier kneedle and coord descent enforce). The shipping path uses both.

The kneedle is a model-selection criterion. The kneedle is not a universal optimum. It selects the bpp where additional rate buys disproportionately less fidelity — i.e., where the marginal cost of further size reduction is high. It is the right shipping criterion under the implicit loss $\mathcal{L}(\text{bpp}, \text{KL}) = \alpha \cdot \text{KL} + \beta \cdot \text{bpp}$ with α, β proportional to the perpendicular axis weights, and a reasonable default in the absence of an explicit task-cost-of-bits curve. Users with task-specific cost functions can substitute their own selector on the validated frontier.

The held-out KL is a calibration-set statistic. The downstream evaluation in Section 7.2 confirms that lower held-out KL does not entail uniform task improvement. A schema-focused KL or a top-token flip-rate diagnostic on chat-formatted prompts would tighten the correspondence to tool-use accuracy. We report the KL we measured and flag the gaps.

Hardware-bounded measurement. The cost hierarchy is calibrated to fit in a 128 GB UMA host; on larger hosts, L_3 candidate budgets and coord-descent pass counts can be substantially larger, and the Pareto sweep can target denser frontiers. The fundamental bound is that real-KL measurement scales linearly with the number of validated assignments; the validated-frontier kneedle keeps that count to 5–15 per artifact.

Limitations. (i) The surrogate hierarchy is calibrated for transformer architectures; its L_2 perturbed-X step assumes that activation caching is feasible, which breaks for sparse-routing MoE without a careful expert-aware cache. (ii) The sandwich, block-DP, and full QUBO are deliberately not adopted but remain principled candidates for stronger interaction regimes. (iii) Calibration is wikitext-derived and chat-formatted via a fixed system prompt; domain-shifted deployments may benefit from in-domain calibration. (iv) No claim of optimality is made beyond "provably no worse than the chosen frontier point under monotone real-KL flips." The frontier itself is sampled, not exhaustive.

9 Conclusion

We described PrismaQuant and its core search algorithm PrismaSCOUT: a mixed-precision LLM quantization pipeline whose defining choice is to use surrogate cost models for candidate generation only and to make every shipping decision on a real, end-to-end Kullback–Leibler divergence measurement against a held-out calibration split. The key engineering challenge was making that measurement loop tractable on a 128 GB UMA host through a multi-level cost hierarchy with lane-batched, tail-amortized L_3, validated-frontier kneedle selection with a noise-floor dominance filter and leave-one-out stability, and a monotone coord-descent polish that is provably non-regressive on real KL.

The shipping artifact on Qwen3.6-27B is 11% smaller and 68% lower in held-out KL than the previously shipped PrismaQuant v1 artifact at the same source. We also documented several principled numerical methods (Lagrangian λ-bisection, sandwich proximal recalibration, block-DP over architectural cliques, sparse pairwise QUBO, top-K Hessian covering, surrogate-only knee, probe-only knee predictor) that were considered during design and either deprecated, made experimental-only, or rejected outright. Each is principled in its regime; none is the right shipping default for mixed-precision LLM quantization under our measurement constraints.

Reproduction. The Qwen3.6-27B PrismaSCOUT artifact is available at https://huggingface.co/rdtand/Qwen3.6-27B-PrismaSCOUT-Blackwell-NVFP4-BF16-vllm (DOI `10.57967/hf/8656`, [1]). The pipeline is open source at https://github.com/RobTand/prismaquant; commits referenced throughout (`8e503a5`, `2e670b3`, `444bc28`, ...) are visible there.

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A Fused-sibling aggregation

Fused siblings in attention (q, k, v) and MoE (gate_up, down) must share one format choice in vLLM's fused kernel. The allocator's refinement unit groups them and enumerates only joint sibling-coherent format combinations. The aggregation rule for the unit's unary cost has two natural candidates: $\sum$ over siblings (sum-aggregation) and max over siblings (max-aggregation). Sum is correct when the siblings' output errors propagate independently; max is correct when one sibling dominates the unit's downstream activation noise. Empirically on Qwen3.6-27B, sum-aggregation matches the measured L_3 end-KL more closely on both attention and MoE units (commit `d370c3e`, "allocator: revert to sum-aggregation for fused siblings (right thing on principle)"). We use sum.

B Allocator runtime parameters

Defaults below are taken from the argument parser of `prismaquant.iterate_perturbed_allocation` at the audit-consensus commit (`2e670b3`).

- L_2: `bit_precision` = 0.001, `convergence-frac` (weighted-Hamming flip ratio) = 0.01, `max-iters` = 5, `ema-decay` = 0.5, `n-calib-samples` = 8, `input-rows` = 256.
- L_3: `l3-uncertainty-rel-tol` = 0.10, `l3-min-fraction` = 0.05, `l3-max-fraction` = 0.30, `l3-safety-fraction` = 0.02, `l3-max-lanes-per-batch` = 64 (UMA-cap aware), `l3-tail-only` = on, `l3-regression-tolerance` = 0.0.
- Validated-frontier kneedle: `DEFAULT_KNEE_KL_NOISE_FLOOR` $\eta = 1 \times 10^{-5}$, bisection tolerance `knee-tolerance` = 0.05 bpp, leave-one-out diagnostic engaged at ≥ 4 frontier points, instability fallback policy `best-kl` (replaces the geometric knee with the lowest-KL frontier point).
- Coord descent: `max-passes` = 1, `early-stop-streak` = 50, `max-lanes-per-batch` = 64.

C Reproduction manifest for the Qwen3.6-27B PrismaSCOUT artifact

The following pointers are sufficient to retrace the production 27B run end-to-end. The artifact itself is at DOI `10.57967/hf/8656`, [1].

- **Pipeline source:** https://github.com/RobTand/prismaquant, audit-consensus commit 2e670b3 (*Implement PR-Polish-1: Rename validation_kl and secure surrogate knee*). The kneedle search, archive DP, and L3 polish reside in prismaquant/iterate_perturbed_allocation.py; L2 perturbed-X cache in prismaquant/perturbed_x_cache.py; propagated-cost L3 in prismaquant/propagated_cost.py; additive multi-choice knapsack DP in prismaquant/allocator_solver.py.
- **Source weights:** Qwen/Qwen3.6-27B (BF16, HF Hub), snapshot taken locally as /hfcache/qwen36-27b-bf16.
- **Calibration:** wikitext-derived prompts, chat-formatted under a fixed system template, hashed by prismaquant.allocator.calibration_data_hash. Splits: l3_calib_split=train (cost generation), kl_calib_split=validation (real-KL gate); selection-time KL evaluated on 8×512 tokens of the validation split.
- **Run directories.** The no-seed archive search lives under /home/rob/dq-runs/qwen3p6-27b-smrf-overnight/ as 27b-knee-archive-beam4-refine-noseed-20260504T022551Z; the export run that produced the shipped checkpoint is in the same parent as 27b-export-5p3095-kneedle-20260504T025719Z. Matched-flag vLLM eval reports (validate_5p3095.md, validate_5p5.md, tool-eval-bench summaries) live under that export run's evals/ subtree.
- **Hardware:** NVIDIA GB10 Blackwell, 128 GB UMA; CUDA toolkit matched to the vLLM container (vllm-fresh-b12x:latest).
- **Serving:** vLLM with --quantization compressed-tensors --kv-cache-dtype fp8 --enable-prefix-caching --max-model-len 32768 --tool-call-parser qwen3_coder.
- **Determinism:** the run is reproducible up to floating-point determinism on the same Blackwell host. The full audit trail (validated-frontier candidates, dominance filter parameters, knee, leave-one-out shadow knees, coord-descent flip log, L2 iteration trace) is written to the run's out/ directory.